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Characterization of the cellar mud odor compounds and their content variation in strong aroma base Baijiu by GC/MS/O and means of molecular sensory science

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ABSTRACT

The aim of this work was to elucidate the specificity of the cellar mud odor compounds in strong aroma base Baijiu using gas chromatography-olfactometry-mass spectrometry analysis based on chemical and sensory profiles. Fifty-three aroma compounds in both base and commercial Baijiu were identified and quantified by comparing their MS, retention indices with those of pure standards. Odor activity values and principal component analysis were used to account for the variation in the sensory attributes of these aroma compounds. The results demonstrate that high acid and phenol contents contribute to the cellar mud aroma profiles in base strong aroma Baijiu. This approach may provide valuable information for product quality control of strong aroma Baijiu products.

1. Introduction

Chinese Baijiu contains numerous odorous compounds (odors) that determine its value. During the Baiju manufacturing process, the flavor of base Baijiu is one of the core elements responsible for commercial Baijiu quality. Before commercial Baijiu is sold in the market, it is required to transform the base Baijiu into commercial Baijiu through aging and blending. In order to objectively evaluate the quality and increase the value of Baijiu, it is necessary to characterize the flavors of different grades of Baijiu and identify their corresponding key compounds. Currently, more than 1874 compounds have been reported in soy sauce Baijiu, which is one of the 12 typical Baijiu aroma styles (He et al., 2021; Hong et al., 2020; Li et al., 2019; Liu et al., 2018; Zheng et al., 2016). The remaining 11 typical aroma styles are the strong, light, rice, herb-like, Feng, complex, Chi, Te, sesame, Fuyu, and Laobaigan aroma styles (Li et al., 2019; Zheng et al., 2016). Strong aroma Baijiu (SAB) is one of the most popular traditionally distilled spirits in China, and its volatile composition is complex. Fan et al. first measure 75 volatile compounds in SAB using headspace solid-phase microextraction (HS-SPME) combined with GC–olfactometry dilution analysis (Fan et al., 2019).

Some volatile compounds in SAB may lead to off-odors, such as fury, sour, musty, and cellar mud odors. In particular, cellar mud odor significantly decreases the quality of SAB and causes great economic losses for the Baijiu industry. To evaluate the value and increase the quality of SAB, it is necessary to elucidate the relationship between off-odors and volatile compounds using various techniques. Dong et al. confirmed that p-cresol, hexanoic acid, butyric acid, and 3-methylindole were the key odorants responsible for the mud odor in Luzhou and Gujinggong Baijiu (Dong et al., 2019). He et al. elucidated the specificity of SAB from different regions using two-dimensional gas chromatography–time-of-flight mass spectrometry and descriptive sensory analysis (He et al., 2020). In 2023, Mu et al. suggested that 29 components of commercial SAB are aroma-active compounds (Mu et al., 2023). p-cresol, a metabolite of *Clostridium butyricum*, has also been suggested to contribute to the mud-like odor of fresh Luzhou Laojiao Baijiu. However, very little is known about the compounds responsible for the cellar mud odor between the base and commercial SAB.

We performed gas chromatography/olfactometry (GC/O) combined with gas chromatography-mass spectrometry (GC/MS) analysis and

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calculated odor activity values (OAVs) to investigate the key cellar mud compounds in the base and commercial SAB. In this study, we (1) identified and quantified important aroma compounds in base and commercial SAB and (2) determined key cellar mud compounds using OAVs and aroma recombination experiments.

2. Materials and methods

2.1. Baijiu samples

Four SAB samples, including A1 (base Baijiu sample, 2 years, 70 % ethanol by volume), A2 (base Baijiu sample, 2 years, 70 % ethanol by volume), B1 (commercial Baijiu sample, 9 years, 70 % ethanol by volume), and B2 (commercial Baijiu sample, 5 years, 70 % ethanol by volume) were provided by Shanxi Liulin Liquor Industry Co., Ltd. Baijiu samples were stored at 4 °C until further analysis (Wang et al., 2020).

2.2. Chemical

All chemicals were of analytical reagent (AR) grade and had least 97 % purity. Ethyl formate (109–94–4), ethyl acetate (141–78–6), ethyl 2-methylpropanoate (97–62–1), ethyl butanoate (105–54–4), ethyl 3-methylbutanoate (108–64–5), ethyl pentanoate (539–82–2), 3-methylbutyl butanoate (106–27–4), ethyl hexanoate (123–66–0), hexyl acetate (142–92–7), propyl hexanoate (626–77–7), ethyl heptanoate (106–30–9), ethyl lactate (106–33–2), butyl hexanoate (626–82–4), ethyl octanoate (106–32–1), 3-methylbutyl hexanoate (2198–61–0), hexyl hexanoate (6378–65–0), ethyl decanoate (110–38–3), ethyl succinate (123–25–1), ethyl phenylacetate (101–97–3), phenethyl acetate (103–45–7), ethyl laurate (106–33–2), ethyl myristate (124–06–1), ethyl hexadecanoate (628–97–7), ethyl oleate (111–62–6), ethyl linoleate (544–35–4), 1-butanol (71–36–3), 2-methylbutanol (137–32–6), 3-methylbutanol (123–51–3), 1-pentanol (71–41–0), 1-hexanol (111–27–3), (*2S, 3S*)2,3-butanediol (19132–06–0), meso-2,3-butanediol (5341–95–7), 1,2-propanediol (57–55–6), benzyl alcohol (100–51–6), phenylethyl alcohol (60–12–8), acetaldehyde (75–07–0), propanal (9057–02–7), 2-methyl propanal (78–84–2), 1,1-dimethoxyethane (534–15–6), 3-methylbutanal (590–86–3), benzaldehyde (100–52–7), acetone (67–64–1), 3-hydroxy-2-butanone (513–86–0), acetic acid (64–19–7), butanoic acid (107–92–6), 3-methylbutanoic acid (503–74–2), pentanoic acid (109–52–4), hexanoic acid (142–62–1), heptanoic acid (111–14–8), octanoic acid (124–07–2), p-cresol (106–44–5), 4-ethylphenol (123–0-79), furfural (98–01–1), 2-methyl-2-butanol (internal standard, IS1, 75–85–4), and 4-octanol (IS2, 589–62–8) were obtained from Sigma-Aldrich (Shanghai, China). Linear retention indices (RIs) were built on the DB-WAX (60 m × 250 μm i.d., 0.25 μm film thickness; J&W Scientific, USA) and HP-5 column (30 m × 250 μm i.d., 0.25 μm film thickness; J&W Scientific, USA) using a C6-C40 n-alkane standard (Sigma-Aldrich) (Li et al., 2023; Wang et al., 2023). Ethanol (High Performance Liquid Chromatography, HPLC, 64–1705) and sodium chloride (AR, 7647–14–5) were purchased from the China National Pharmaceutical Group Corp. (Shanghai, China).

2.3. Identification of potential odor compounds

2.3.1. HS-SPME

HS-SPME analysis was performed using a modified version of the method reported by Sun et al. (2022). Baijiu samples (A1, A2, B1, and B2) were diluted with saturated NaCl solution to 10 % ethanol (v/v), and 8.0 mL of the solution was added to 20-mL headspace bottles. The subsequent extraction process was performed as follows: the samples were preheated at 40°C for 15 min, and then the solid-phase microextraction (SPME) fiber (divinylbenzene/carboxen/polydimethylsiloxane; 75 μm film thickness) was exposed over the Baijiu samples headspace and extracted at 45 °C for 40 min. After extraction, the SPME fiber was inserted into the GC inlet for desorption at 250 °C for 5 min. The column effluent was spitless, and helium (99.999 %) was used at a flow rate of 1.5 mL/min.

2.3.2. GC/MS and GC/O for identification of odorants

Gas chromatography/mass spectrometry (GC/MS) was performed to confirm the presence of aromatic compounds in Baijiu. The initial oven temperature was set at 40 °C, held for 2 min, increasing at a rate of 3 °C/min to 180 °C, finally ramped to 240 °C at a rate of 6 °C/min and held for 5 min. The MS was operated in the full scan mode at conditions of mass to charge ratio range and ion source temperature of 35–400 and 150°C. Mass spectra were stored at 10 s intervals. The MS spectra were compared with the MS spectra from the National Institute of Standards and Technology (NIST) and Wiley MS libraries (Sun et al., 2022; Wei et al., 2023).

Gas chromatography/olfactometry (GC/O) analysis was combined with GC/MS analysis. The Sniffer 9,100 Olfactory Detector Port (Gerstel, Germany) was used to confirm the aroma profile of the Baijiu samples. The transmission line temperature was 280 °C, and the carrier gas was helium with a purge valve. Moist nitrogen was used to prevent dehydration of the nasal mucosa of the odor assessors. The detection time was approximately 63.7 min. For each sample, an odor assessor recorded the odor styles, odorants, intensities, and the retention times. A six-point scale ranging from “none” (0) to “very strong” (5) was used for intensity judgment according to Sun’s and Jia’s studies (Jia et al., 2021; Sun et al., 2022).

2.4. Quantification of potential odorants

Three quantification methods were used to obtain accurate results. All experiments were repeated thrice. Except for phenols, compounds including esters, alcohols, aldehydes, and ketones were quantified using GC/FID (Wang et al., 2022). These compounds were pretreated as follows: saturated NaCl solution was added to the Baijiu samples until the ethanol concentration reached 10.0 % to reduce the effect of high ethanol content. Ten μL of the internal standards (IS1 and IS2) were added to each Baijiu sample, as described above. The treated samples were analyzed using GC/FID wtih a DB-WAX column (60 m × 250 μm i.d., 0.25 μm film thickness; J&W Scientific, USA). The samples (1.0 μL) were injected into the above GC in the splitless mode. The initial oven temperature was 40 °C and maintained for 4 min, then increased to 60 °C at a rate of 4 °C/min, then increased to 130 °C at a rate of 6 °C/min, and finally increased to 210°C at a rate of 15 °C/min and maintained for 15 min. The inlet temperature was 240 °C, the carrier gas was nitrogen (99.999 %), and the flow rate was 1.0 mL/min. As described in Section 2.3.2, phenols were analyzed using the HS/SPME/GC/MS method.

Acids were analyzed using an ion chromatography (IC) method (Rune et al., 2023; Dekker et al., 2023). A Dionex ICS-3000 ion chromatograph (Thermo Fisher Scientific) equipped with a silver electrode, an ICS-3000 Dual pump, an AS-AP autosampler, and an ICS-3000 DC detector/chromatography module was used. The run of each Baijiu sample was conducted within 65 minutes using an IonPac^{TM} AS1-HC RFIC^{TM} (4 × 250 mm, 7.5 μm; Thermo Fisher Scientific) analytical column, at an isocratic flow with deoxygenated demineralized H_2O and an EGC KOH Eluent Generator Cartridge. The eluent flow and target concentrations were set at 1.1 mL/min and 10 mM, respectively. The column and compartment temperature were set to 30 and 35°C, respectively. A standard acid solution was prepared to determine acid retention time.

2.5. Aroma profile

Training was performed with 10 panelists (five males and five females, with an average age of 26 years) and lasted for more than one year, especially on regular sensory evaluations (Xu et al., 2022). Additional training was provided to these evaluators to identify and determine the aroma characteristics of the SAB samples. Subsequently, the

evaluators were asked to describe the sensory characteristics of samples A and B (40 mL in a 120 mL ISO standard wine tasting glass). High-frequency words were recorded, and the sensory evaluators agreed on the determination of descriptors for the aroma profile of the SAB samples. Eight aromatic styles were selected to evaluate the aroma profiles of the samples A and B in detail, in which the corresponding reference standards were fruity (ethyl acetate), acidic (a mixture of acetic acid: butanoic acid: pentanoic acid = 25:5:1), alcoholic (ethanol), aldehydic (3-methylbutanal), grain (dried wheat), toasted (3-hydroxy-2-butanone), cellar (furfural), and cellar mud odor (p-cresol). Ten

Table 1
Aroma compounds identified by GC/O/MS in samples A and B.

NO.	Compound	*RI* DB-WAX Hp-5		odor descriptor	Identification [a]	Concentration (mg/L) [b]	
						A [c]	B [c]
	esters						
1	ethyl formate	—	605	fruity	RI, MS, aroma, S	1.10±0.01	43.62±0.98
2	ethyl acetate	—	656	sweet, fruity	RI, MS, aroma, S	573.12±2.11	1046.34±2.58
3	ethyl 2-methylpropanoate	—	759	ester	RI, MS, aroma, S	6.79±0.28	8.79±0.48
4	ethyl butanoate	989	804	sweet, fruity	RI, MS, aroma, S	70.16±0.19	159.28±0.16
5	ethyl 3-methylbutanoate	1042	858	sweet, fruity	RI, MS, aroma, S	3.37±0.06	5.58±0.56
6	ethyl pentanoate	1132	905	fruity	RI, MS, aroma, S	3.60±0.09	46.27±0.06
7	3-methylbutyl butanoate	1226	—	fruity	RI, MS, aroma, S	2.56±0.02	3.56±0.03
8	ethyl hexanoate	1246	1020	fruity	RI, MS, aroma, S	45.79±1.44	1754.25±2.34
9	hexyl acetate	1275	1025	fruity	RI, MS, aroma, S	8.07±0.23	5.07±0.13
10	propyl hexanoate	1317	1099	fruity	RI, MS, aroma, S	5.49±0.03	3.49±0.02
11	ethyl heptanoate	1337	1104	pineapple	RI, MS, aroma, S	4.57±0.57	25.55±0.52
12	ethyl lactate	1348	—	fruity	RI, MS, aroma, S	1399.49±61.67	739.69±23.55
13	butyl hexanoate	1414	1195	grape	RI, MS, aroma, S	12.82±0.72	2.19±0.12
14	ethyl octanoate	1438	1203	fruity, fatty	RI, MS, aroma, S	27.62±0.51	37.62±0.51
15	3-methylbutyl hexanoate	1537	1299	green	RI, MS, aroma, S	0.89±0.07	2.81±0.15
16	hexyl hexanoate	1610	1389	fruity	RI, MS, aroma, S	6.16±0.31	4.16±0.21
17	ethyl decanoate	1638	1398	fruity, grape	RI, MS, aroma, S	0.52±0.07	0.92 0.21
18	ethyl succinate	1692	—	sweet	RI, MS, aroma, S	1.74±0.85	1.85±0.36
19	ethyl phenylacetate	1782	1249	fruity, sweet	RI, MS, aroma, S	4.83±0.29	0.93±1.32
20	phenethyl acetate	1804	—	floral	RI, MS, aroma, S	9.13±0.74	3.48±2.85
21	ethyl laurate	1916	—	green grass	RI, MS, aroma, S	1.87±0.19	1.17±0.16
22	ethyl myristate	2123	—	medicinal	RI, MS, aroma, S	1.77±0.50	0.23±0.02
23	ethyl hexadecanoate	2259	1996	waxy	RI, MS, aroma, S	15.08±1.28	10.59±0.13
24	ethyl oleate	2479	2168	fatty, waxy	RI, MS, aroma, S	4.47±0.50	3.19±0.44
25	ethyl linoleate	2529	2162	—	RI, MS, aroma, S	11.54±0.29	2.22±0.03
	alcohols						
1	1-butanol	1158	—	malty	RI, MS, aroma, S	55.83±1.05	81.84±2.09
2	2-methylbutanol	1201	—	fruity	RI, MS, aroma, S	5.02±0.32	49.62±0.44
3	3-methylbutanol	1217	732	malt	RI, MS, aroma, S	29.25±0.87	203.65±3.90
4	1-pentanol	1259	—	fruity	RI, MS, aroma, S	4.70±0.69	5.84±0.45
5	1-hexanol	1361	879	flower, fruity	RI, MS, aroma, S	37.82±0.86	52.38±1.21
6	(2*S*, 3*S*)2,3-butanediol	1526	994	green	RI, MS, aroma, S	65.06±7.14	30.46±9.53
7	meso-2,3-butanediol	1538	1077	mushroom	RI, MS, aroma, S	33.55±3.08	10.48±1.97
8	1,2-propanediol	1564	1080	fruity	RI, MS, aroma, S	101.76±10.98	34.10±5.18
9	benzyl alcohol	1864	—	fruity	RI, MS, aroma, S	0.54±0.08	nd
10	phenylethyl alcohol	1909	1117	rose, honey	RI, MS, aroma, S	15.30±1.60	5.65±0.75
	aldehydes						
1	acetaldehyde	1391	—	aldehydic	RI, MS, aroma, S	58.85±5.86	296.45±14.27
2	propanal	—	792	sweaty	RI, MS, aroma, S	0.51±0.07	1.84±0.30
3	2-methyl propanal	1427	—	floral	RI, MS, aroma, S	2.56±0.24	5.89±0.40
4	1,1-dimethoxyethane	1515	—	green	RI, MS, aroma, S	1.32±0.09	278.37±0.28
5	3-methylbutanal	1533	1163	cucumber	RI, MS, aroma, S	7.51±3.02	27.88±4.03
6	benzaldehyde	1633	1049	fruity	RI, MS, aroma, S	2.27±0.38	16.17±2.03
	ketones						
1	acetone	—	827	—	RI, MS, aroma, S	1.97±0.31	10.74±0.08
2	3-hydroxy-2-butanone	1689	1330	mushroom	RI, MS, aroma, S	0.57±0.01	1.57±0.02
	acids						
1	acetic acid	1452	701	vinegar	RI, MS, aroma, S	791.02±20.98	486.02±23.55
2	butanoic acid	1628	—	sweaty	RI, MS, aroma, S	66.71±2.07	25.85±1.21
3	3-methylbutanoic acid	1672	895	sweaty	RI, MS, aroma, S	5.38±0.85	3.17±0.12
4	pentanoic acid	1739	983	sweaty	RI, MS, aroma, S	610.56±18.19	314.47±13.53
5	hexanoic acid	1849	1090	sweaty	RI, MS, aroma, S	272.64±10.29	411.82±20.51
6	heptanoic acid	1956	1147	sweaty	RI, MS, aroma, S	141.12±9.74	15.45±1.02
7	octanoic acid	2061	1231	sweaty	RI, MS, aroma, S	64.88±0.08	12.29±1.51
	phenols						
1	p-cresol	2079	1083	cellar mud odor	RI, MS, aroma, S	11.45±1.60	3.90±0.53
2	4-ethylphenol	2172	1175	cellar mud odor	RI, MS, aroma, S	11.20±0.80	13.47±1.97
	others						
1	furfural	1461	836	sweet, almond	RI, MS, aroma, S	202.97±10.10	47.93±1.51

[a] MS, aroma compounds were identified by MS; RI, linear retention indices, the retention indices of the compounds on DB-WAX and HP-5 columns were calculated based on the peak time of n-alkanes (C6–C30) on the instrument; S, compounds were identified by pure standards; aroma, the odors were compared with the standard compounds by GC/O/MS.

[b] Concentrations of aroma compounds were expressed as the mean value of two samples ± standard deviation (mean ± SD).

[c] Data A was averaged for samples A1 and A2. Data B was averaged for samples B1 and B2.

sensory evaluators were asked to evaluate 40 mL of sample A or B, which were added to an ISO standard wine-tasting glass. The intensities of the eight different odors were determined as follows: 0 (none), 1 (extremely weak), 2 (weak), 3 (middle), 4 (strong), and 5(very strong). Evaluation experiments were performed in a sensory panel room at 20 ± 1 °C with a humidity of 45–50 %. Three parallel sessions were conducted for each sample.

2.6. OAVs

To assess the contribution of each aroma compound to the overall aroma profile of SAB, the odor activity value (OAV) was calculated as follows:

$$OAV = C_{aroma} \times T_{odor}{}^{-1}$$

where C_{aroma} is the aroma concentration (μg L^{-1}) measured in the Baijiu and T_{odor} is its odor threshold (μg L^{-1}).

2.7. Data analysis

All the experiments were performed in triplicate. Principal component analysis (PCA) was applied to cluster both the base and commercial SAB samples in terms of their aroma profile (Zhang et al., 2023). Partial least squares discriminant analysis (PLS-DA) was used to maximize the separation among samples and to build relationships between the compositional variables (x data) and sensory styles (y data) in the base and commercial SAB samples. PCA and PLS-DA regression analyses were performed using SIMCA version 14 software (Umetrics, Umearing, Sweden). The concentrations of the aroma compounds were calculated using SPSS 25.0. After significant effects were identified, differences between means were considered significant at 0.05 based on Tukey's honestly significant difference (Tukey HSD) test. Results were expressed as the mean value of two samples ± standard deviation (mean ± SD). Differences were considered statistically significant with one asterisk at $p < 0.05$.

3. Results and discussion

3.1. Concentrations of aroma compounds in Samples A and B

To identify the odor compounds that contributed to the overall aroma profile, samples A and B were analyzed using GC/O and HS/SPME/GC/MS. These compounds encompass a range of VOCs, including esters, alcohols, aldehydes, ketones, acids, phenols, and furans (Xu et al., 2022). A total of 53 odorants (Table 1) were identified, including 25 esters, 10 alcohols, six aldehydes, two ketones, seven acids, two phenols, and one furan. The identities of these compounds were confirmed using mass spectrometry, aroma, RIs, and pure standards. As shown in Table 1, the RI values of these 53 aroma compounds ranged from 605 to 2529, according to the GC/MS analysis.

As shown in Table 1, the quantitative results indicated that acids, esters, and alcohols were relatively abundant. Among those aroma compounds, the contents of eight aroma compounds (ethyl lactate, acetic acid, pentanoic acid, ethyl acetate, hexanoic acid, furfural, heptanoic acid, and 1,2-propanediol) with contents above 100 mg/L were quantified in sample A. However, 12 aroma compounds (ethyl hexanoate, ethyl acetate, ethyl lactate, acetic acid, hexanoic acid, pentanoic acid, acetaldehyde, 1,1-dimethoxyethane, 3-methylbutanol, ethyl butanoate, heptanoic acid, and octanoic acid) with contents above 100 mg/L were quantified in sample B. This is consistent with the observation that the SAB is rich in ester and acid substances (Wu et al., 2022). Among the phenols, the content of p-cresol in sample A was above 10 mg/L and higher than that in sample B. Phenols are found in different types of liquors and red wine brands and are known to account for different flavors and bitterness due to their interaction with salivary glycoproteins (Alara et al., 2021). In terms of aldehydes and ketones, the contributions of both aldehydes and ketones in sample A were lower than those in B. Furan and its derivatives may be formed by carbohydrate dehydration. Subsequently, both cyclization in Maillard-type reactions and accumulation during the later stages of wine aging contributed to the formation of furan and its derivatives.

As shown previously by Dong et al., 3-methylindole (3-MI) was detected using GC/MS with 4.4–142.8 μg/L in base SAB and contributed to cellar mud odor (Dong et al., 2018). In their reports, the odor threshold of 3-MI in the 46 % ethanol-water solution was determined to be 6.9 μg/L. However, it was not detected using GC/MS in samples A and B. It may be due to the different geographical origin of base Baijiu. Li et al. found that 4-ethylphenol was detected with cellar mud odor in light aroma Baijiu, which was quietly similar with the odor of the 'animal'-like (Li et al., 2023).

3.2. OAVs

The overall aroma profile obtained from the ten evaluators indicated that sample A exhibited stronger cellar mud odor compared with sample B. The contents of aroma compounds and their thresholds impact the odor activity (Song et al., 2020). As shown in Table 2, the ratio of the content of each compound to the corresponding odor threshold (odor activity value (OAV)) was used to evaluate the contribution of each aroma compound to the overall aroma profile of SAB. A total of 34 and 36 aroma compounds with OAV ≥ 1 were found in samples A and B, respectively.

The compounds with the highest OAV were pentanoic acid (1569) in sample A and ethyl hexanoate (31722) in sample B, followed by ethyl butanoate (861, 1954 in samples A and B, respectively). In addition to the above compounds, ethyl pentanoate and 3-methylbutanal showed higher OAVs (≥ 1000) in sample B, which were considered the skeleton components of SAB (Mu et al., 2023). The pentanoic acid, butanoic acid, p-cresol, butyl hexanoate, ethyl phenylacetate, ethyl lactate, phenethyl acetate, 3-methylbutanoic acid, acetic acid, furfural, hexyl hexanoate, hexyl acetate and phenylethyl alcohol concentrations were 1.5–5.9 times higher in sample A than those in B. To date, few studies have evaluated the contributions of acids and phenols to the odor of SAB. These 13 aromatic compounds may be the most important odorants in samples A and B. p-cresol, with a strong cellar mud odor, had an OAV of 69 in sample A. However, the content was extremely low (11.45 mg/L).

Ethyl hexanoate exhibited the highest OAV in sample B, which was in line with the reports by Huang et al. (2018) and Zheng et al. (2016). It was remarkable that ethyl hexanoate contributed to the major aroma in SAB and sesame flavor Baijiu. Huang et al. found that OAV of p-cresol in sesame flavor-type Baijiu was 5, which was smaller than that in sample A. Niu et al. reported that the OAV values of acetic acid and 3-methylbutyric acid in Laimao baijiu were much higher than those of other acids, indicating that the aroma of Laimao baijiu was different with SAB (Niu et al., 2020).

Through OAV analysis, acids and p-cresol were confirmed to be cellar mud odor markers. In the process of fermentation, acids react with ethanol to form esters through enzyme catalysis. Once that the highest esterification rate is reached, the esters will decompose into alcohols and acids, which plays a major role in the formation of SAB flavor (Niu et al., 2020). This result emphasizes that fermentation process needs to be considered for future developments. Therefore, it is necessary to provide good living environment for the microorganisms. Firstly, sufficient water should be supplied with water/grain mass ratio 60 %-80 %. Secondly, the grain temperature should be controlled in the process of entering cellar, precisely. In summer, it is the same as the ground temperature. In winter, it should be between 18.0- 22.0 °C to prevent yeast aging. Finally, the cellar should be sealed to prevent the loss of moisture and volatile aroma compounds.

Table 2
Odor threshold and OAVs of 53 aroma compounds in samples A and B.

NO.	Compound	odor threshold (μg/L) [a]	Average OAVs [c]	
			A [d]	B [d]
1	pentanoic acid	389.1	1569	808
2	ethyl butanoate	81.5	861	1954
3	ethyl hexanoate	55.3	828	31722
4	ethyl 3-methylbutanoate	6.9	488	809
5	3-methylbutanal	16.51	455	1689
6	ethyl pentanoate	26.8	134	1726
7	ethyl 2-methylpropanoate	54.5	125	161
8	hexanoic acid	2520	108	163
9	butanoic acid	964.64	69	27
10	p-cresol	167	69	23
11	acetaldehyde	1200	49	247
12	octanoic acid	2700	24	5
13	1-butanol	2730	20	30
14	butyl hexanoate	678	19	3
15	ethyl acetate	32550	18	32
16	ethyl phenylacetate	407	12	2
17	ethyl lactate	128080	11	6
18	heptanoic acid	13820	10	1
19	phenethyl acetate	908.83	10	4
20	4-ethylphenol	1230.1	9	11
21	1-hexanol	5370	7	10
22	3-methylbutanoic acid	1050	5	3
23	acetic acid	160000	5	3
24	Furfural	44030	5	1
25	hexyl hexanoate	1890	3	2
26	3-methylbutyl butanoate	915	3	4
27	3-hydroxy-2-butanone	259	2	6
28	ethyl octanoate	12900	2	3
29	2-methyl propanal	1300	2	5
30	hexyl acetate	5560	1	1
31	3-methylbutyl hexanoate	1400	1	2
32	1,1-dimethoxyethane	2090	1	133
33	benzaldehyde	4200	1	4
34	phenylethyl alcohol	28920	1	<1
35	ethyl decanoate	1120	<1	1
36	propyl hexanoate	12780	<1	<1
37	ethyl heptanoate	13150	<1	2
38	3-methylbutanol	179190	<1	1
39	1-pentanol	37400	<1	<1
40	2,3-butanediol	668000	<1	<1
41	2,3-butanediol	668000	<1	<1
42	2-methylbutanol	211800	<1	<1
43	benzyl alcohol	40930	<1	<1
44	ethyl laurate	494000	<1	<1
45	ethyl lactate	128080.0	<1	<1
46	phenylethyl alcohol	28920.0	<1	<1
47	3-methylbutanol	179190.0	<1	<1
48	1-pentanol	37400.0	<1	<1
49	ethyl formate	—[b]		
50	propanal	—[b]		
51	ethyl hexadecanoate	—[b]		
52	ethyl oleate	—[b]		
53	ethyl linoleate	—[b]		

[a] Odor thresholds were determined in 46 % alcohol by volume from reference 16.
[b] Odor thresholds was unavailable.
[c] OAVs were calculated by concentration by the odor thresholds. Average OAVs, the average odor activity values of two samples with the same flavor Baijiu.
[d] Data A was averaged for samples A1 and A2. Data B was averaged for samples B1 and B2.

3.3. Relationships between the sensory attributes and aroma compounds

The radar chart showed (Fig. 1A) that the base and commercial SAB had significant differences in all aroma attributes. The intensities of cellar mud and acidic odors were significantly different ($p < 0.01$). However, samples A and B exhibited similar intensities of roasted and aldehydic attributes. The PCA model was used to determine the relationship between the different aroma attributes and samples. It is suggested that A1, A2, B1, and B2 were clearly separated from the score plot, with PC1 and PC2 explaining the total variance of 84.82 % and 11.42 %, respectively (Fig. 1B). The base SAB samples (A1 and A2) are well -differentiated on the left side and positively correlated with the cellar, cellar mud, and acidic descriptors. The commercial SAB samples (B1 and B2) were well -distributed on the right side and were positively correlated with the grain, toasted, aldehydic, fruity, and alcoholic descriptors.

As one of the key aroma compounds in SAB, 1-butanol contributes an alcoholic odor to the overall aroma (He et al., 2020). To discriminate SAB from different regions, 1-butanol was selected as the marker compound. The fruity aroma is correlated with esters, which primarily originate from fermentation and distillation processes (Gao et al., 2014). Among the esters in SAB, ethyl acetate, ethyl hexanoate, ethyl pentanoate, and ethyl butanoate have been reported to have high OAVs and are the key aroma compounds in Baijiu. The pyrazines, ketones such as 3-penten-2-one and phenylacetone, and 2,4,5-trimethyloxazole contribute to the "toasted" aroma. Among these compounds, pyrazines are products of the Maillard reaction between sugars and amino acids, and are positively correlated with coconut, roasted, and caramel-like aromas (Fan et al., 2007). Some furanoids, such as furfural, 5-methyl furfural, 2-pentanoylfuran, and 2-acetylfuran are related to the "cellar" aroma, together with 2-propionylfuran, 5-methyl-2--phenyl-2-hexenal, and 2-methyl-6-ethylpyrazine (He et al., 2020). Furanoids are hypothesised to interact with other aroma compounds to contribute the "cellar" aroma. The "grain" aroma is associated with the fermentation and distillation processes of sorghum, rice, wheat and other grains. Ketones and esters, such as ethyl cyclohexanoate, 2-pentanone, 3-hydroxy-2-butanone, and isoamyl hexanoate are not thought to contribute to the "grain" aroma individually; however, the combination of the above compounds are thought to contribute the "grain" aroma. The "acidic" aroma is positively correlated with acetic acid, butanoic acid, and 3-methylbutanoic acid (Ferreira et al., 2016). The cellar mud aroma is associated with pentanoic acid, hexanoic acid, heptanoic acid, and p-cresol. p-cresol is reported as a special kind of phenol, which possesses a cellar mud aroma with a low odor threshold (167 μg/L) even at a low concentration (Mu et al., 2023).

4. Confirmation of the key compounds related to the cellar mud odor

The OPLS-DA model was used to further identify the profiles of each pair of samples. Fig. 2A showed that most esters, p-cresol, heptanoic acid, octanoic acid and furfural were associated with samples A1 and A2. Most alcohols, aldehydes, and ketones were correlated with samples B1 and B2. Differential compounds were identified using variable importance in projection (VIP) scores (>1.0) (Fig. 2B). Ten flavor compounds with VIP > 1 were screened using statistical analysis: ethyl hexanoate, ethyl pentanoate, 3-methylbutanal, ethyl butanoate, pentanoic acid, ethyl 3-methylbutanoate, acetaldehyde, 1,1-dimethoxyethane, hexanoic acid, and p-cresol. Among the differential compounds identified in sample A, ethyl hexanoate, ethyl pentanoate, 3-methylbutanal, ethyl butanoate, pentanoic acid, ethyl 3-methylbutanoate, p-cresol, and acetaldehyde scored from 1.033 to 3.101 with OAVs ranging from 49 to 1569. However, 1,1-dimethoxyethane had a VIP score of 1.400 with a lower OAV (1) due to a higher odor threshold (2090 μg/L). In contrast, ethyl hexanoate, which had a lower content (45.79 mg/L) in sample A, showed the highest VIP score of 3.10. Based on the OPLS-DA model, some compounds had higher OAV with lower VIP, such as ethyl 2-methylpropanoate (125) and butanoic acid (69) with VIP scores of 0.95 and 0.96, respectively.

Fig. 3. shows the average aroma content of acids and p-cresol in samples A and B, which tended to increase the cellar mud odor intensity. The acid and p-cresol contents in sample A was 1.9–9.1 times than those in sample B. Among these aroma compounds, p-cresol is a potent off-flavor compound which causes a cellar mud odor (Mu et al., 2023). It

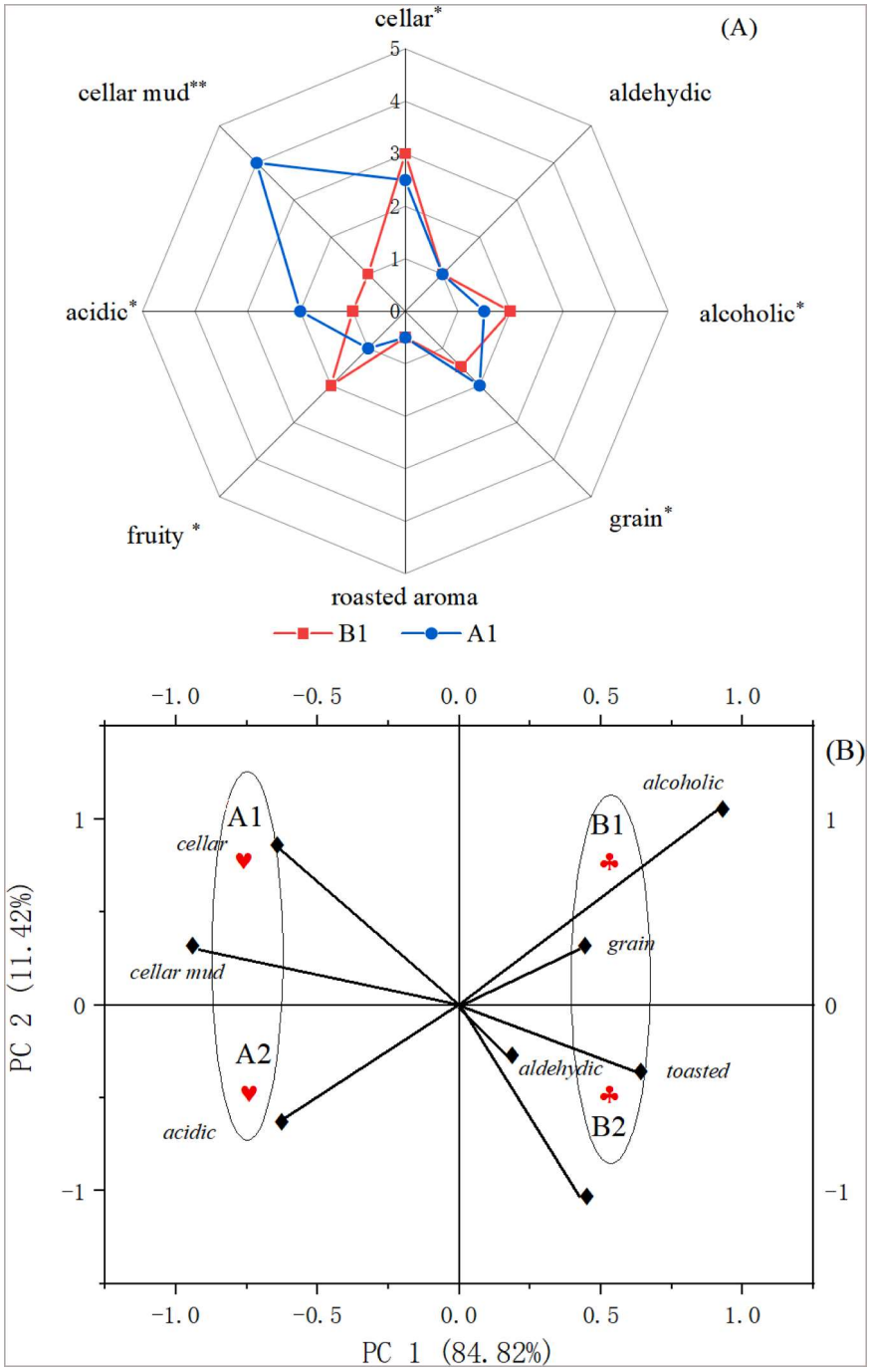


Fig. 1. (A) Aroma profile of the mean sensory scores of the base and commercial samples. Notations: Number indicates a significant aroma difference among the three cups via the triangle test based on 10 panelists. * and ** indicate the significance of the difference between sample means at $P<0.05$ and $P<0.01$, respectively. (B) PCA score plot based on the sensory scores.

has been reported that pentanoic acid has a low threshold of 389.1 g/L and always causes a sweaty and cellar mud odor in Baijiu. Altogether, these results indicate that cellar mud odor formation was mainly due to an increase in certain odorants in the Baijiu samples.

To verify the obtained analytical data, a total of **12** compounds with higher contents in sample A were added in the corresponding amounts to commercial sample B. As shown in Table 3, a comparison of the aroma profiles of the spiked sample A to sample B showed that the cellar mud odor could be simulated and was enhanced relative to that in sample B. Simultaneously, the aldehydic, alcoholic, and fruity aromas were partially masked, and their intensities decreased compared to those in sample B. Grain and toasted aromas did not present significant differences between spiked sample A and sample B, indicating that the cellar mud odor was successfully simulated by the addition of the identified constituents.

Furthermore, this study provides an innovative information by investigating and validating the off-odor of SAB. It offers target compounds for the off-odor removal process. In previous studies, activated carbon and resin were used to remove the off-odor of Baijiu (Liu et al., 2024). A limitation of this process is that these materials can also remove important flavor compounds, affecting sensory attributes of Baijiu. Materials must be further developed to have advantages in high off-odor removal rate and low aroma loss rate. Based on this study, molecularly imprinting may be a powerful approach for the preparation of materials

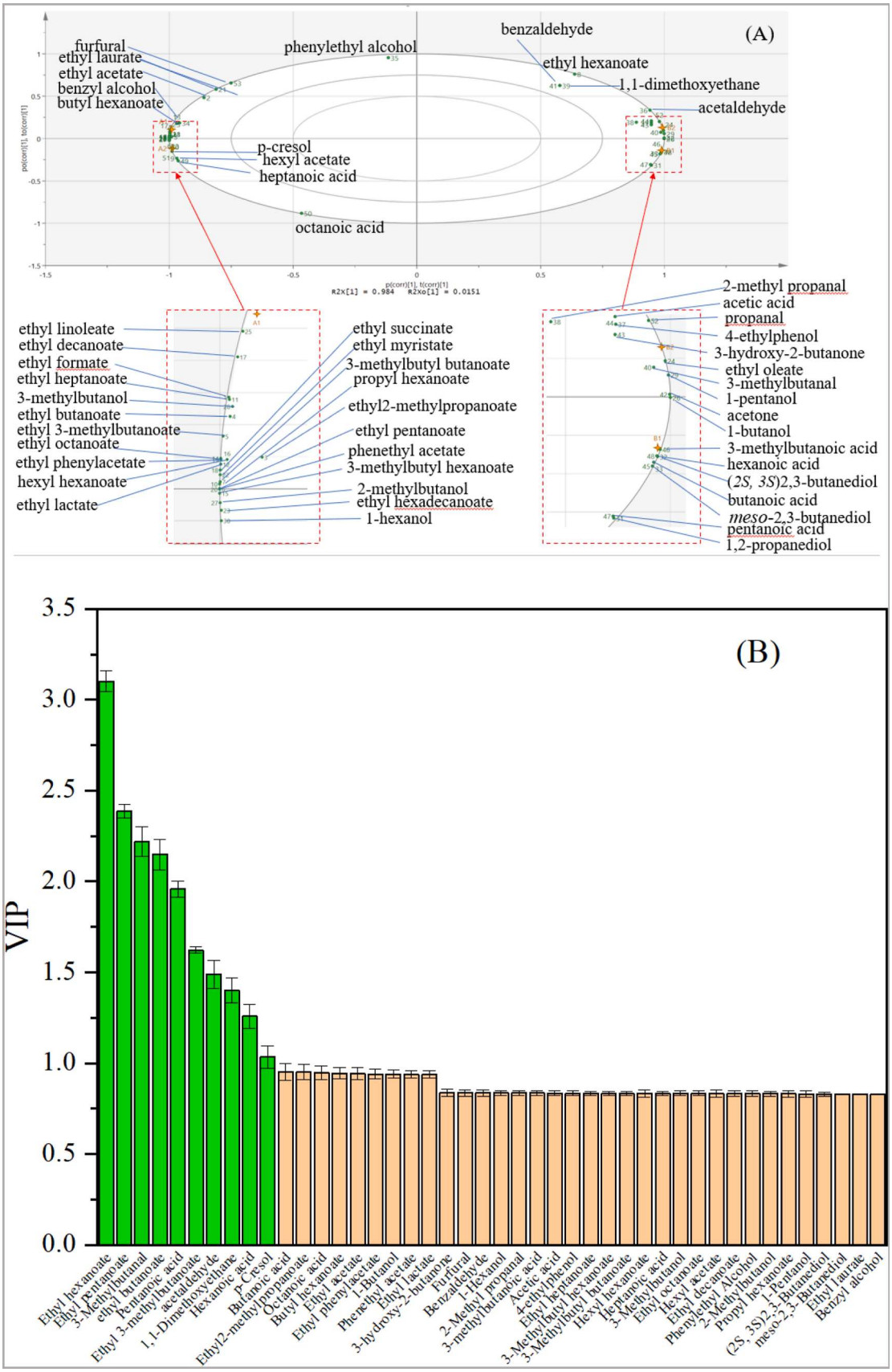


Fig. 2. (A) PLS-DA analysis based on aroma active compounds. (B) The VIP values were calculated from the PLS regression model.

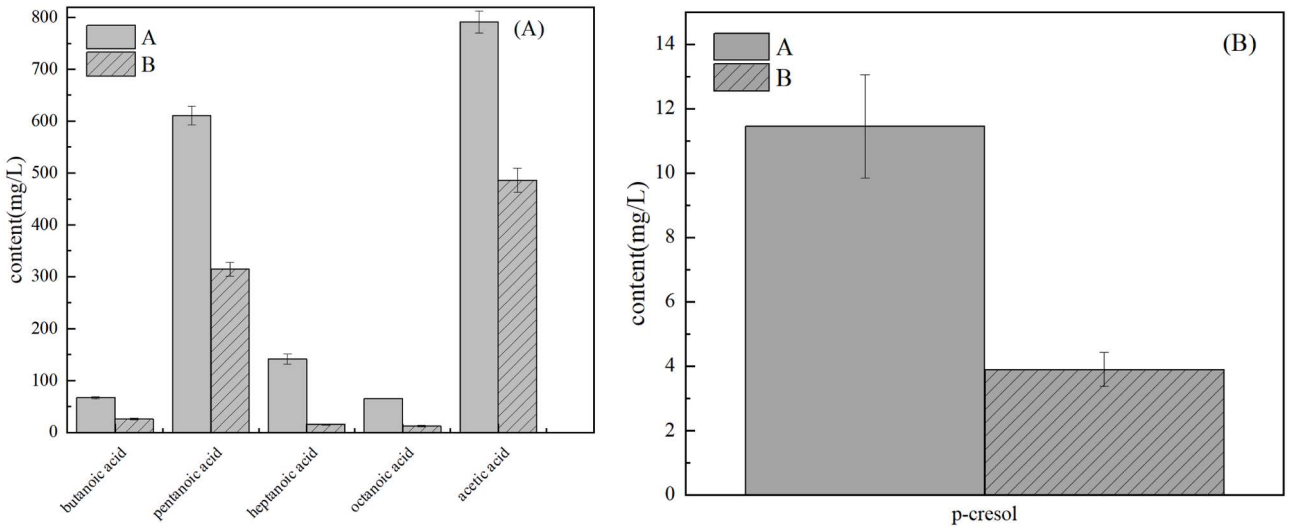


Fig. 3. Average concentrations of some aroma compounds in samples A and B. Notations: Bars indicate standard errors. Average concentrations, the average concentrations of two samples with the same flavor Baijiu.

Table 3
Addition tests to commercial SAB.

No	aroma compounds added to the B sample	odor quality	n/10 [a]	significance [b]
1	acetic acid	vinegar	8/10	***
	butanoic acid	sweaty,		
	pentanoic acid	sweaty		
	heptanoic acid	sweaty		
	octanoic acid	sweaty		
2	p-cresol	cellar mud	9/10	***
3	butyl hexanoate	fruity	3/10	ns
	ethyl phenylacetate	sweet		
	ethyl lactate	fruity floral		
	phenethyl acetate	fruity		
	hexyl hexanoate			
4	Furfural	sweet almond	2/10	ns

[a] Number indicates a significant aroma difference among the three cups via the triangle test based on 10 panelists.
[b] *** ($P < 0.001$) indicate the significance of the difference between sample means.

having acids or phenols recognition properties.

4. Conclusions

In conclusion, compounds causing cellar mud odor in base and commercial SAB were successfully identified using both GC/MS/O analysis and sensory evaluation methods. The aroma profile of the cellar mud odor was successfully recreated in spiked sample. It was confirmed that high acidic compounds and p-cresol contents were the key contributors to the cellar mud odor. When these compounds were added to the corresponding concentrations, there was a significant positive growth trend in the cellar mud odor intensity.

Author Contributions

Conceptualization, W.Z.; methodology, X.H. and D.W.; validation, W.J.; formal analysis, L.H. and Y.C.; resources, F.Z., J.H. and X.M.; writing—original draft preparation, W.Z.; writing—review and editing, X.H.; All authors have read and agreed to the published version of the manuscript.

Ethics statement

We declare sensory evaluations in this work don't require ethical approval in our country. All procedures were reviewed and approved by China National Research Institute of Food and Fermentation Industries. A verbal informed consent was obtained from each subject prior to their participation in the evaluation of the samples. All subjects participated voluntarily and have the right to withdraw from this research at any time. The privacy rights of human subjects have always been observed.

The authors declare that no experiments were conducted on animals in this research.

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Data availability

Not applicable

Conflicts of Interest

The authors declare no conflict of interest

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.afres.2024.100671.

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